

The empirical recurrence for OEIS A253331: recovering a conjecture that is not written down

Adrian Perez Fontelles
Independent researcher

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Abstract

OEIS A253331 records “Empirical recurrence of order 65”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 1482$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 65, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 65 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A253331 is “Number of $(6+1)X(n+1)$ 0..1 arrays with every $2X2$ subblock sum nondecreasing horizontally, vertically and antidiagonally ne-to-sw.”. It has offset 1 and begins

1482, 4218, 11795, 34660, 98741, 283804, 797324, 2204476, ...

The entry states:

Empirical recurrence of order 65 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 13:58:23, revision 6) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 1482$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 1482$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 2964$ exact terms of the model returns order 65 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{65} c_i a(n-i),$$

c_1	4	c_{18}	−83472604	c_{35}	−16608067010	c_{52}	376838000
c_2	34	c_{19}	−464626628	c_{36}	32350308542	c_{53}	−73929632
c_3	−154	c_{20}	321617740	c_{37}	13392143806	c_{54}	−102098304
c_4	−538	c_{21}	1203607388	c_{38}	−30656527454	c_{55}	24365376
c_5	2854	c_{22}	−994217596	c_{39}	−9035226298	c_{56}	22125024
c_6	5178	c_{23}	−2684402036	c_{40}	25395914600	c_{57}	−6130368
c_7	−33906	c_{24}	2557363928	c_{41}	4905527998	c_{58}	−3687168
c_8	−32736	c_{25}	5180989660	c_{42}	−18345271546	c_{59}	1150464
c_9	290118	c_{26}	−5584624388	c_{43}	−1938383006	c_{60}	443648
c_{10}	129822	c_{27}	−8681178604	c_{44}	11510659258	c_{61}	−152576
c_{11}	−1904406	c_{28}	10482191396	c_{45}	339821090	c_{62}	−34304
c_{12}	−177982	c_{29}	12646376212	c_{46}	−6238287538	c_{63}	12800
c_{13}	9974314	c_{30}	−17046679540	c_{47}	232726186	c_{64}	1280
c_{14}	−1679162	c_{31}	−16009545692	c_{48}	2898087941	c_{65}	−512
c_{15}	−42798958	c_{32}	24143960269	c_{49}	−279407618		
c_{16}	16013807	c_{33}	17565246464	c_{50}	−1142348144		
c_{17}	153297886	c_{34}	−29876521462	c_{51}	170254784		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 65, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $65 + 4$ terms removed returns order 65 as well, so $\deg q_{\min} = 65 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 22 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $65 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 65 that this sequence satisfies — and that family turns out to have exactly one member.

References

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